

Pedro Miguel Sánchez Sánchez

Data Scientist and Researcher — AI, Federated Learning, Cybersecurity & Distributed Systems

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About Me

I am currently a Data Scientist at Roche, where I apply advanced AI and data-driven methods to solve real-world problems in healthcare and life sciences. Previously, I earned a Ph.D. in Computer Science with research on IoT cybersecurity using Machine Learning and Deep Learning for device behavior fingerprinting and attack detection, in collaboration with armasuisse S&T, Switzerland. Afterward, I advanced my work into Federated Learning and AI Trustworthiness, extending these methods to IoT, foundation models, and trustworthy AI. To stay at the cutting edge, I continue pursuing applied training in areas such as Computer Vision, NLP, and AI agents.

Experience

- Data Scientist, Roche** *Madrid, Spain Sept 2025 – ...*
 - AI application in different business cases
- Senior Researcher, Funditec (Avantix Technology Foundation)** *Madrid, Spain Nov 2024 – Sept 2025*
 - Applied ML/DL and Cybersecurity, European and National research project preparation and execution
- Postdoctoral Researcher, University of Murcia** *Murcia, Spain Feb 2024 – Nov 2024*
 - New methods in Federated Learning and trustworthy AI assessment → armasuisse S&T collaboration
 - Contributed to EU-GUARDIAN EDF and ROBUST-6G Horizon projects
- Visiting Researcher, Cyber-Defence Campus, armasuisse S&T** *Thun, Switzerland Sept–Dec 2021, 2022*
 - Visiting researcher for six months (two three-month periods) in the context of my PhD thesis
- Predoctoral Researcher, University of Murcia** *Murcia, Spain Sept 2019 – Feb 2024*
 - Worked on IoT device behavior modeling for identification and attack detection, funded by armasuisse S&T

Education

- Ph.D. in Computer Science, Cum Laude. University of Murcia** *Sept 2019 – Feb 2024*
 - Thesis: *Identical IoT device identification via hardware performance fingerprinting and Machine Learning*
 - Supported by *armasuisse S&T*, also involved in the EU-GUARDIAN EDF and 5GZORRO H2020 projects
- M.Sc. in Computer Science, 9.07/10. University of Murcia** *Sept 2018 – June 2019*
 - Thesis: *Securing Smart Offices through an Intelligent and Multi-device Continuous Authentication System*
- B.Sc. in Computer Science, 8.04/10. University of Murcia** *Sept 2014 – June 2018*
 - Graduated with honors. Networking Specialization. Thesis: *Continuous authentication for mobile devices*

Research

18 h-index (+20 JCR journals, +10 confs, +4 chapters). Relevant research on Federated Learning, applied ML/DL in IoT behavior security, and AI Trustworthiness. Updated publication list in [Google Scholar](#) [🔗](#).

Relevant Projects (Grants and Code)

- [NEBULA](#) [🔗](#) Decentralized Federated Learning Platform → Research, design and implementation advice
- [EU-GUARDIAN EDF](#) [🔗](#), Cyber defence automation → Research, architecture design, and implementation
- [ROBUST-6G](#) [🔗](#) and [5G-ZORRO](#) [🔗](#) Horizon projects on distributed AI for networks → Research and design

Awards and Achievements

- National Young Researcher Award in Computer Science** *2024*
 - Granted by the Scientific Society of Computer Science of Spain and the BBVA Foundation [🔗](#)
- Outstanding PhD Thesis Award** *2024*
 - Award for the best Computer Science PhD Thesis in 2024 at my university (University of Murcia)
- Elsevier Computer Networks Best Paper Award** *2024*
 - "Federated learning for malware detection in IoT devices", a collab with EPFL, UZH, and CYD Campus
- Conference Travel Grant and Volunteering** *2022, 2023*
 - Student grants for conference attendance/volunteering: AAAI'23, ACM SIGCOMM'23, IEEE NOMS'22
- Secure and Private AI Scholarship Nanodegree** *2019*
 - One of the 5000 recipients of 2019 Udacity's Secure and Private AI Scholarship sponsored by *Facebook*. Topics covered: Federated Learning, Differential Privacy, Homomorphic Encryption.

Technical Skills

Specialties: Federated Learning, AI Trustworthiness, Machine/Deep Learning, Cybersecurity, Behavior, IoT
Programming and Frameworks: Python, Bash, Keras, PyTorch, scikit-learn, C/C++, Java

Annex. Main Publications

- [1] **Sánchez Sánchez, Pedro Miguel**, E. T. Martínez Beltrán, M. Fernández Llamas, G. Bovet, G. Martínez Pérez, and A. Huertas Celdrán, “Profe: Communication-efficient decentralized federated learning via distillation and prototypes,” in *Proceedings of the IEEE International Conference on Communications (IEEE ICC)*. IEEE, 2025, p. Accepted.
- [2] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, “Transfer learning in pre-trained large language models for malware detection based on system calls,” in *MILCOM 2024-2024 IEEE Military Communications Conference (MILCOM)*. IEEE, 2024, pp. 853–858.
- [3] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, N. Xie, G. Bovet, G. Martínez Pérez, and B. Stiller, “Federatedtrust: A solution for trustworthy federated learning,” *Future Generation Computer Systems*, vol. 152, pp. 83–98, 2024.
- [4] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, “Adversarial attacks and defenses on ml-and hardware-based iot device fingerprinting and identification,” *Future Generation Computer Systems*, vol. 152, pp. 30–42, 2024.
- [5] A. Huertas Celdrán, **Sánchez Sánchez, Pedro Miguel**, J. Von Der Assen, T. Schenk, G. Bovet, G. Martínez Pérez, and B. Stiller, “Rl and fingerprinting to select moving target defense mechanisms for zero-day attacks in iot,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 5520–5529, 2024.
- [6] C. Feng, A. Huertas Celdrán, J. Baltensperger, E. T. Martínez Beltrán, **Sánchez Sánchez, Pedro Miguel**, G. Bovet, and B. Stiller, “Sentinel: An aggregation function to secure decentralized federated learning,” in *European Conference on Artificial Intelligence (ECAI) 2024*. IOS Press, 2024, pp. 1760–1767.
- [7] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, G. Bovet, and G. Martínez Pérez, “Single-board device individual authentication based on hardware performance and autoencoder transformer models,” *Computers & Security*, vol. 137, p. 103596, 2024.
- [8] **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, T. Schenk, A. L. B. Iten, G. Bovet, G. Martínez Pérez, and B. Stiller, “Studying the robustness of anti-adversarial federated learning models detecting cyberattacks in iot spectrum sensors,” *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 2, pp. 573–584, 2024.
- [9] E. T. Martínez Beltrán, **Sánchez Sánchez, Pedro Miguel**, S. López Bernal, G. Bovet, M. Pérez Gil, G. Martínez Pérez, and A. Huertas Celdrán, “Fedstellar: A platform for training models in a privacy-preserving and decentralized fashion,” in *International Joint Conference on Artificial Intelligence (IJCAI-23) Demos*, 2023.
- [10] A. Huertas Celdrán, J. Kreischer, M. Demirci, J. Leupp, **Sánchez Sánchez, Pedro Miguel**, M. Franco Figueredo, G. Bovet, G. Martínez Pérez, and B. Stiller, “A framework quantifying trustworthiness of supervised machine and deep learning models,” in *SafeAI2023: The AAAI’s Workshop on Artificial Intelligence Safety*, 2023, pp. 2938–2948.
- [11] E. T. Martínez Beltrán, M. Pérez Quiles, **Sánchez Sánchez, Pedro Miguel**, S. López Bernal, G. Bovet, M. Pérez Gil, G. Martínez Pérez, and A. Huertas Celdrán, “Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges,” *IEEE Communications Surveys & Tutorials*, vol. 25, no. 4, p. 2983–3013, 2023.
- [12] V. Rey, **Sánchez Sánchez, Pedro Miguel**, A. Huertas Celdrán, and G. Bovet, “Federated learning for malware detection in iot devices,” *Computer Networks*, vol. 204, p. 108693, 2022.
- [13] **Sánchez Sánchez, Pedro Miguel**, J. M. Jorquera Valero, A. Huertas Celdrán, G. Bovet, M. Pérez Gil, and G. Martínez Pérez, “A survey on device behavior fingerprinting: Data sources, techniques, application scenarios, and datasets,” *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1048–1077, 2021.